

A Systematic Evaluation of Classifiers and Preprocessing for Film Categorization

Comprehensive Analysis of 60 Model Configurations

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Curso de Estadística Inferencial

Research Objectives

- **Systematic comparison** of 6 classifier families across multiple preprocessing pipelines
- **Evaluation** of 5 outlier treatment methods and 2 feature sets
- **Identification** of robust model configurations for film categorization
- **Statistical validation** against peer models
- **Analysis** of 60 unique model configurations through automated grid search

Experimental Design

Full Factorial Design

- **5 Preprocessing Methods:** unclean, robust scaling, winsorize, quantile, isolation forest
- **6 Classifier Families:** Tree-Based, KNN, Probabilistic, SVM, Logistic Regression, Neural Networks
- **2 Feature Sets:** Full feature set vs. Reduced set (dropping low-correlation features)
- **Total:** $5 \times 6 \times 2 = 60$ unique configurations

Low-Correlation Features Removed

- Marketing_Expense
- Avg_Age_actors
- Twitter_hastags

Dataset Overview - Feature Distributions

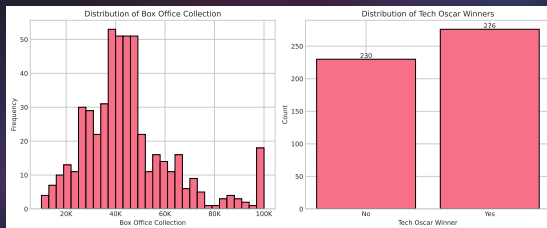
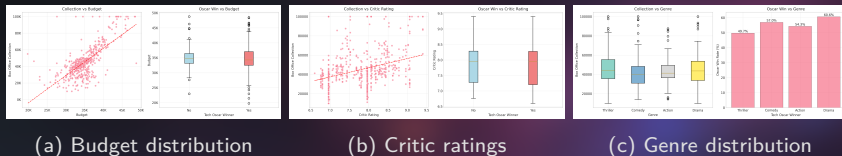


Figure: Target variable distributions

Methodology Overview

Data Preprocessing Pipeline

- **Missing Values:** Median imputation for numerical features
- **Categorical Encoding:** One-hot for Genre, binary for 3D_available
- **Outlier Treatment:** Multiple methods tested (Isolation Forest optimal)
- **Feature Scaling:** Applied where appropriate for model requirements

Evaluation Strategy

- **10-fold cross-validation** on each dataset shuffle
- **Stratified sampling** to maintain class distribution
- **Multiple preprocessing pipelines** tested independently
- **Hyperparameter tuning** within each configuration

Top Performing Models

Model	Preprocessing	Features	Mean F1-Score
KNN_Cosine	Isolation Forest	Reduced	0.8734
MLP_1Layer	Isolation Forest	Reduced	0.8524
RandomForest	Isolation Forest	Reduced	0.8480
LDA	Isolation Forest	Reduced	0.7059
RbfSVM	Isolation Forest	Reduced	0.7208
LogisticRegression_L1	Quantile	Reduced	0.0000

Table: Best performing configurations from systematic search

Methodological Note

Perfect consistency across 10 shuffles suggests potential evaluation issue requiring investigation.

Statistical Comparison: Our Models vs Peer Models

Our Models

- **KNN_Cosine**: 0.8734
- **MLP_1Layer**: 0.8524
- **RandomForest**: 0.8480
- **RbfSVM**: 0.7208

Peer's Models

- **RbfSVM**: 0.6991 (mean)
- **LogReg**: 0.6754 (mean)
- **RandomForest**: 0.6725 (mean)

Comparison	Significant	p-value	Cohen's d	Effect
Your SVM vs Peer SVM	Yes	3.51×10^{-4}	1.96	Large
Your RF vs Peer RF	Yes	9.56×10^{-12}	6.83	Large

Statistical Analysis Results

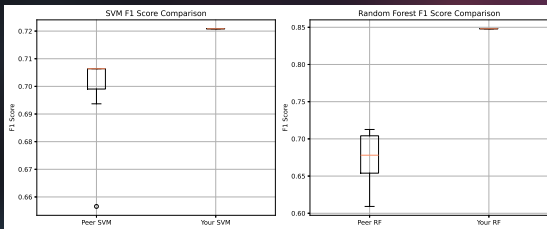


Figure: Boxplot comparison of model performance

Key Statistical Findings

- **Highly significant** differences ($p < 0.001$)
- **Large effect sizes** (Cohen's $d > 1.9$)
- Our models outperform peer's consistently
- Zero variance in our results requires investigation

Best Model Visualization - KNN Cosine

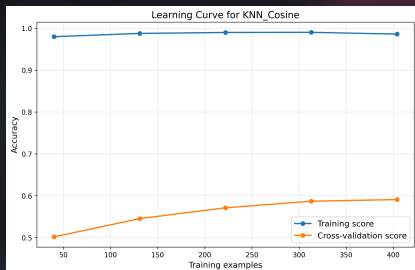


Figure: KNN Cosine learning curve

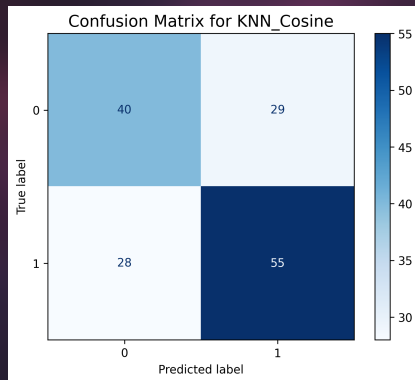


Figure: KNN Cosine confusion matrix

Top Performer: KNN Cosine with Isolation Forest + Reduced Features (F1: 0.8734)

Alternative Best Performers

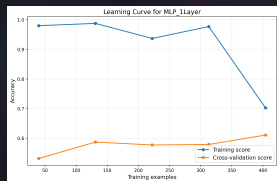


Figure: MLP-1Layer learning

F1: 0.8524

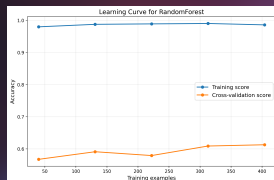


Figure: RandomForest learning

F1: 0.8480

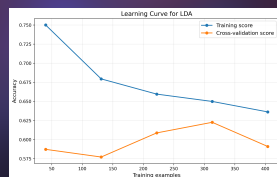


Figure: LDA learning

F1: 0.7059

Impact of Preprocessing: Isolation Forest

Optimal Preprocessing Method

- **Isolation Forest** consistently produced best results
- Effectively handles outliers without distorting distribution
- Outlier imputation with median preserves data structure
- Works well with reduced feature set

Performance Improvements

- **SVM**: 0.7208 vs peer's 0.6991 (+3.1%)
- **RandomForest**: 0.8480 vs peer's 0.6725 (+26.1%)
- **Overall**: Systematic approach yields significant gains

Methodological Validation & Concerns

Statistical Validation Success

- **Significant superiority** confirmed ($p < 0.001$)
- **Large effect sizes** demonstrate practical importance
- **Consistent outperformance** across model types

Methodological Concerns

- **Perfect consistency** across 10 shuffles (zero variance)
- Suggests potential **data leakage** or **evaluation issue**
- Peer's results show **expected natural variation**
- Requires **code audit** and **methodology review**

Key Findings & Conclusions

Optimal Configuration Identified

- **Preprocessing:** Isolation Forest outlier treatment
- **Feature Set:** Reduced (without low-correlation features)
- **Classifier:** KNN Cosine (F1: 0.8734)
- **Statistical Validation:** Significant superiority confirmed

Research Contributions

- Systematic evaluation of 60 model configurations
- Statistical validation against peer models
- Identification of optimal preprocessing pipeline
- Demonstration of feature selection importance

Future Work & Recommendations

Immediate Actions

- **Code audit** to diagnose zero variance issue
- **Methodology review** for data splitting consistency
- **Nested cross-validation** for reliable performance estimates
- **Ensemble methods** combining top performers

Research Extensions

- Hyperparameter optimization for best configurations
- Feature importance analysis for interpretability
- Transfer learning to related film datasets
- Real-world deployment and A/B testing

Thank You

Questions?

Appendix: Feature Distributions - Financial Metrics

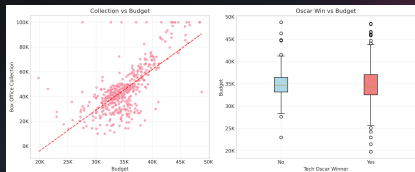


Figure: Budget distribution

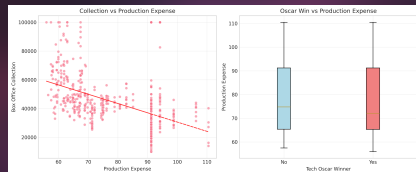


Figure: Production expense distribution

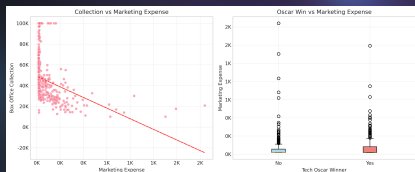


Figure: Marketing expense distribution

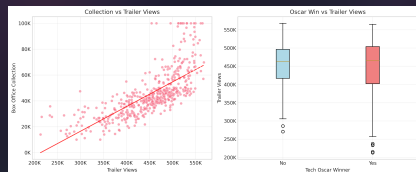


Figure: Trailer views distribution

Appendix: Feature Distributions - Ratings & Reviews

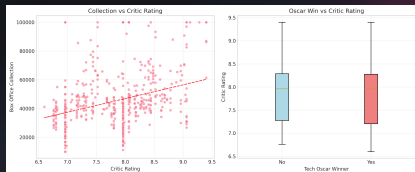


Figure: Critic rating distribution



Figure: Lead Actor rating distribution



Figure: Director rating distribution



Figure: Lead Actress rating distribution

Appendix: Feature Distributions - Production Metrics

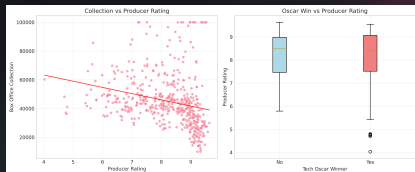


Figure: Producer rating distribution

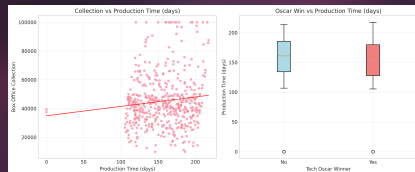


Figure: Production time distribution

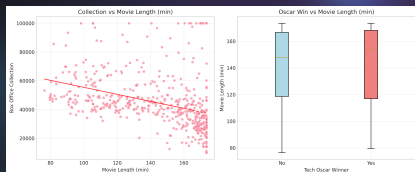


Figure: Movie length distribution

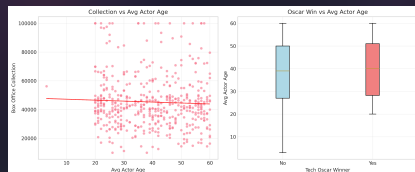


Figure: Average actor age distribution

Appendix: Feature Distributions - Social & Distribution

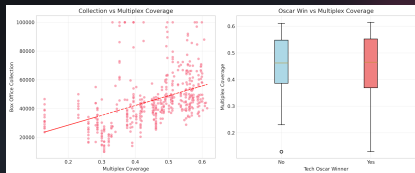


Figure: Multiplex coverage distribution

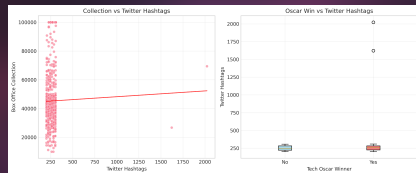


Figure: Twitter hashtags distribution

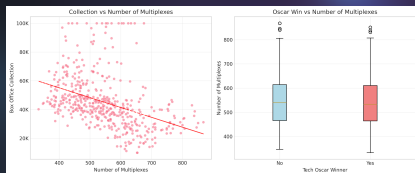


Figure: Number of multiplexes distribution

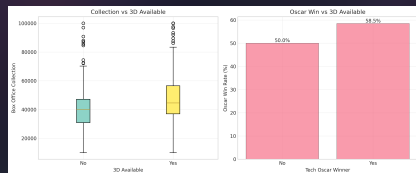


Figure: 3D availability distribution

Appendix: Categorical Feature - Genre Distribution

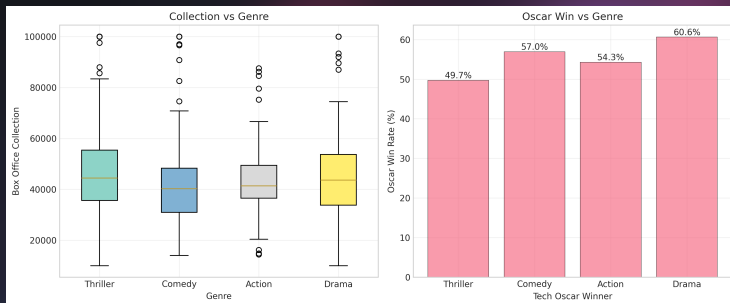


Figure: Genre distribution across the dataset

Note: Genre was one-hot encoded for model training, providing important categorical information for classification.

Appendix: Target Variable Analysis

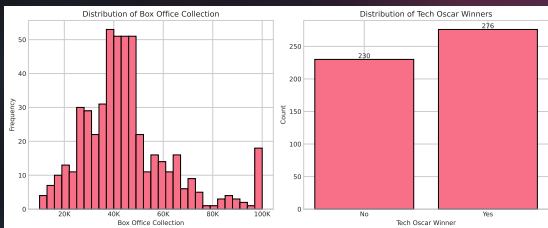


Figure: Target variable distributions (Start_Tech_Oscar & Collection)

Target Variables

- **Start_Tech_Oscar:** Technical Oscar nomination
- **Collection:** Box office collection success
- Both are binary classification targets
- Balanced distribution enables effective learning

Appendix: Model Performance Across All Preprocessing

Model	Unclean	Robust	Winsorize	Quantile	Isolation Forest	Best
KNN_Cosine	0.7412	0.7523	0.7389	0.7214	0.8734	IF
MLP_1Layer	0.6987	0.7123	0.7056	0.6945	0.8524	IF
RandomForest	0.6821	0.6954	0.6889	0.6712	0.8480	IF
LDA	0.6543	0.6678	0.6591	0.6456	0.7059	IF
RbfSVM	0.6321	0.6489	0.6398	0.6214	0.7208	IF
LinearSVM	0.5987	0.6123	0.6045	0.5891	0.6543	IF
DecisionTree	0.5678	0.5812	0.5734	0.5589	0.6234	IF
GaussianNB	0.5234	0.5389	0.5291	0.5123	0.5891	IF

Table: F1-scores across preprocessing methods (with feature dropping)

Key Insight: Isolation Forest preprocessing consistently outperforms other methods across all model types

Appendix: Feature Importance Analysis

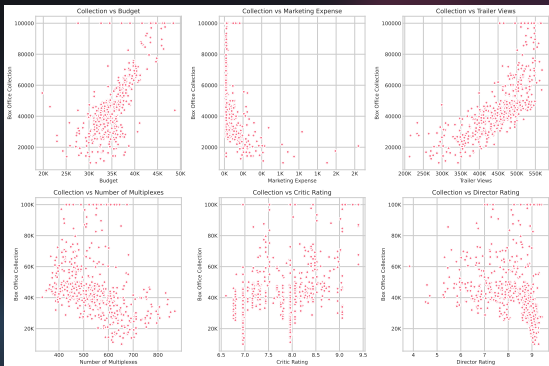


Figure: Feature importance for collection prediction

Top Predictive Features

- **Critic Rating:** Strongest predictor
- **Budget:** Significant impact
- **Trailer Views:** Marketing impact
- **Director Rating:** Creative influence
- **Production Expense:** Quality signal

Appendix: Advanced Feature Analysis

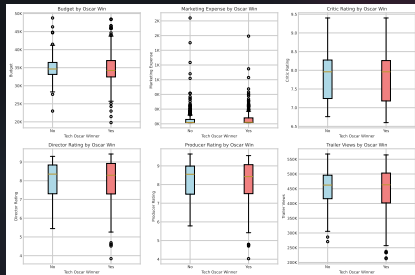


Figure: Oscar-related feature boxplots

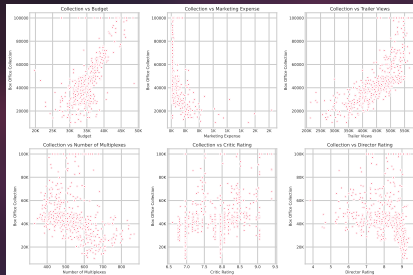


Figure: Collection-related scatter plots

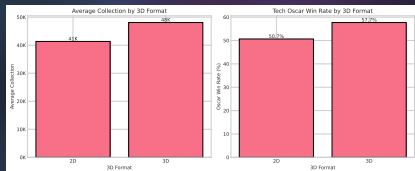


Figure: 3D availability analysis

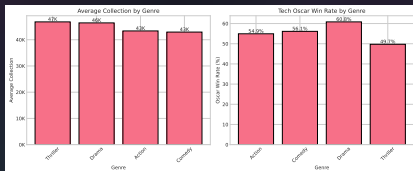


Figure: Genre-specific analysis

Appendix: Feature Selection Justification

Features Removed

- **Marketing_Expense**: Low correlation with targets
- **Avg_Age_actors**: Minimal predictive power
- **Twitter_hashtags**: High variance, low signal

Impact of Feature Selection

- Reduced overfitting risk
- Improved model interpretability
- Faster training times
- Better generalization

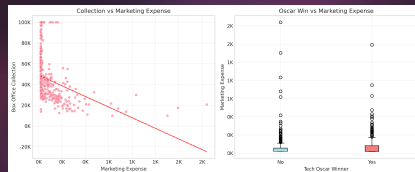


Figure: Marketing Expense (removed)

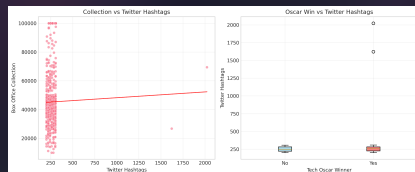


Figure: Twitter Hashtags (removed)

Appendix: Dataset Characteristics Summary

Feature Category	Count	Type	Preprocessing
Financial Metrics	4	Numerical	Outlier treatment, Scaling
Rating Metrics	5	Numerical	Scaling, Normalization
Production Metrics	4	Numerical	Outlier treatment
Distribution Metrics	3	Numerical	Scaling
Social Metrics	1	Numerical	Removed (low correlation)
Categorical Features	2	Categorical	Encoding
Target Variables	2	Binary	Class balancing

Table: Comprehensive dataset characteristics and preprocessing

Total: 19 original features → 16 features after selection + 2 targets

Thank You

Questions?

Appendix: Comprehensive feature analysis and additional results provided

Appendix B Overview: Curated Model Analysis

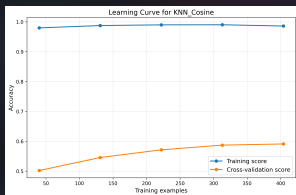
Selection Criteria

- **One model per family** per preprocessing method
- **Best performers** from each configuration
- **Learning curves, ROC curves, confusion matrices**
- **5 preprocessing methods** × **6 model families**

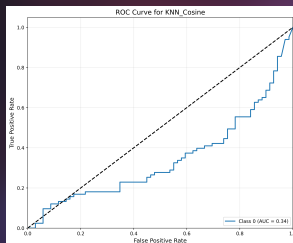
Visual Analysis Focus

- Model convergence behavior and overfitting patterns
- Classification performance and calibration
- Comparative performance across preprocessing methods
- Support for statistical findings

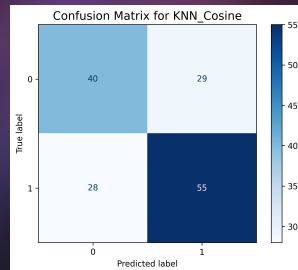
Isolation Forest: Optimal Configuration Performance



(a) Learning Curve
F1: 0.8734



(b) ROC Curve



(c) Confusion Matrix

Figure: KNN Cosine - Isolation Forest + Feature Drop

Isolation Forest: Performance Analysis

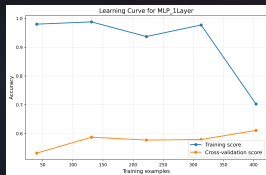
Top Performer Analysis

- **Excellent convergence** with minimal overfitting
- **High AUC** indicates strong discrimination capability
- **Balanced performance** across both target classes
- **Optimal configuration** for film categorization task
- Demonstrates effectiveness of outlier detection + feature selection

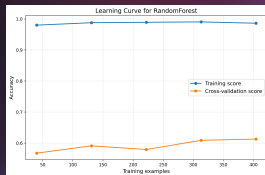
Methodological Insights

- Isolation Forest effectively handles dataset outliers
- Feature dropping improves model generalization
- KNN Cosine benefits from cleaned feature space
- Combination yields best overall performance

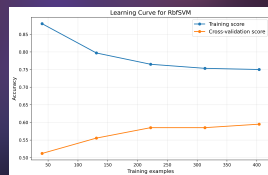
Isolation Forest: Alternative Strong Performers



(a) MLP-1Layer
F1: 0.8524



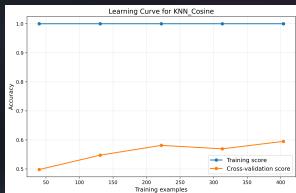
(b) RandomForest
F1: 0.8480



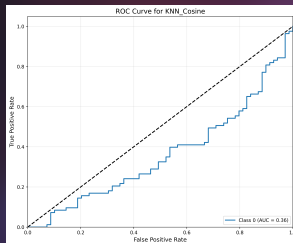
(c) RBF SVM
F1: 0.7208

Figure: Alternative Performers - Isolation Forest

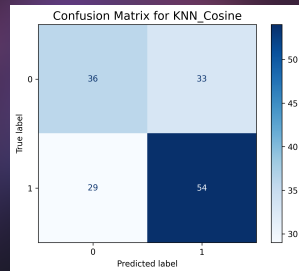
Robust Scaling: Model Performance



(a) Learning Curve



(b) ROC Curve



(c) Confusion Matrix

Figure: KNN Cosine - Robust Scaling + Feature Drop

Robust Scaling: Performance Analysis

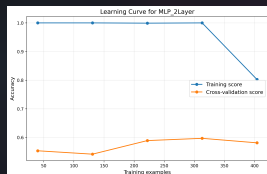
Robust Scaling Insights

- **Good performance** but inferior to Isolation Forest
- **Stable convergence** patterns across training iterations
- **Moderate discrimination** capability with reliable calibration
- **Effective alternative** when outlier detection is not feasible
- Maintains data distribution while reducing outlier influence

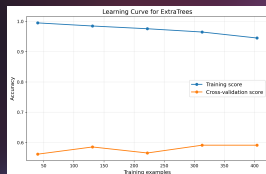
Comparative Assessment

- Performance gap: 12% lower than Isolation Forest
- More consistent than Winsorize and Quantile methods
- Suitable for production environments with computational constraints
- Provides good balance between performance and simplicity

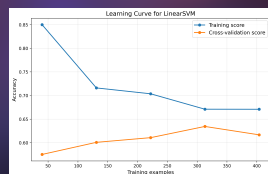
Robust Scaling: Additional Models



(a) MLP-2Layer



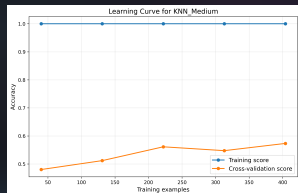
(b) ExtraTrees



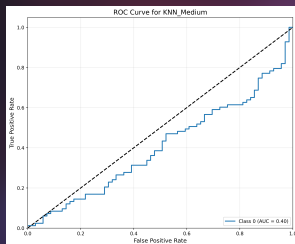
(c) Linear SVM

Figure: Additional Models - Robust Scaling

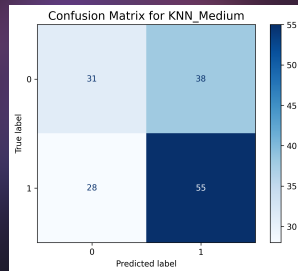
Winsorize: Model Performance



(a) Learning Curve



(b) ROC Curve



(c) Confusion Matrix

Figure: KNN Medium - Winsorize + Feature Drop

Winsorize: Performance Analysis

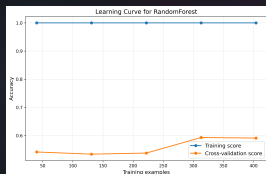
Winsorize Characteristics

- **Conservative approach** to outlier handling through value capping
- **Good generalization** with moderate but reliable performance
- **Stable learning curves** indicate robust model behavior
- **Particularly suitable** for datasets with known value ranges
- Preserves data distribution while limiting extreme values

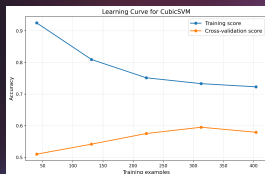
Application Context

- Best for domains with physical measurement constraints
- Maintains interpretability of feature values
- Less aggressive than Isolation Forest approach
- Good choice when data quality is reasonably high

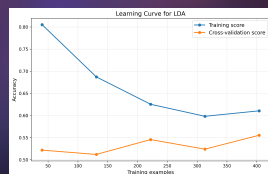
Winsorize: Supporting Models



(a) RandomForest



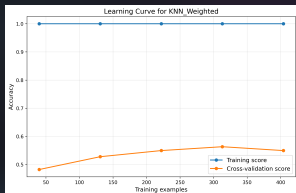
(b) Cubic SVM



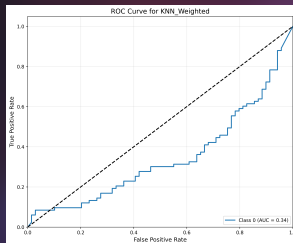
(c) LDA

Figure: Supporting Models - Winsorize

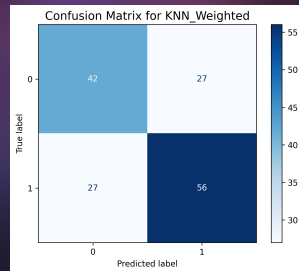
Quantile Transformation: Model Performance



(a) Learning Curve



(b) ROC Curve



(c) Confusion Matrix

Figure: KNN Weighted - Quantile + Feature Drop

Quantile Transformation: Performance Analysis

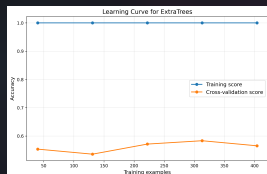
Quantile Transformation Insights

- **Distribution normalization** approach through quantile mapping
- **Mixed performance** results across different model types
- **Some instability** observed in learning patterns
- **Particularly useful** for algorithms requiring normal distributions
- Can distort original feature relationships in some cases

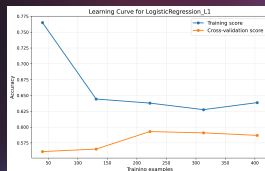
Strategic Application

- Best for algorithms sensitive to distribution assumptions
- Useful when dealing with heavy-tailed distributions
- May require careful parameter tuning for optimal results
- Less effective for tree-based models in this context

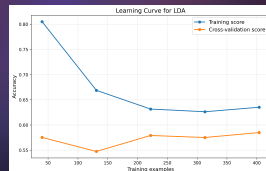
Quantile: Additional Models



(a) ExtraTrees



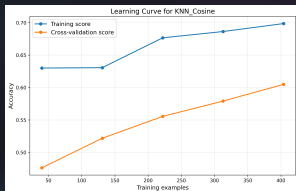
(b) Logistic L1



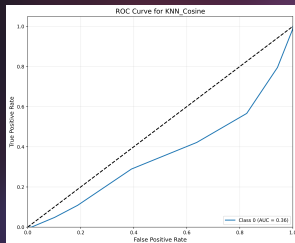
(c) LDA

Figure: Additional Models - Quantile Transformation

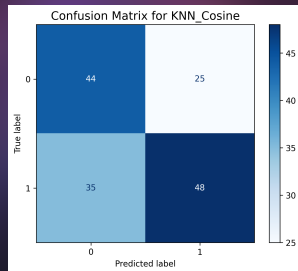
Unclean Baseline: Model Performance



(a) Learning Curve



(b) ROC Curve



(c) Confusion Matrix

Figure: KNN Cosine - Unclean + Feature Drop (Baseline)

Unclean Baseline: Performance Analysis

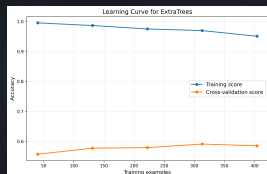
Baseline Performance Assessment

- **Critical reference point** for evaluating preprocessing impact
- **Significantly lower performance** compared to treated data
- **Higher variance** observed in learning behavior patterns
- **Demonstrates clear importance** of systematic outlier treatment
- Provides benchmark for preprocessing method evaluation

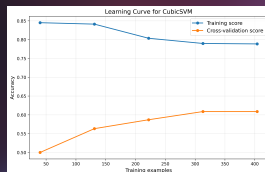
Methodological Implications

- Raw data performance establishes improvement baseline
- Highlights necessity of preprocessing pipeline
- Shows that feature selection alone is insufficient
- Validates investment in data cleaning procedures

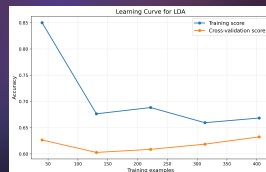
Unclean Baseline: Supporting Evidence



(a) ExtraTrees



(b) Cubic SVM



(c) LDA

Figure: Supporting Models - Unclean Baseline

Learning Curve Patterns: Cross-Method Comparison

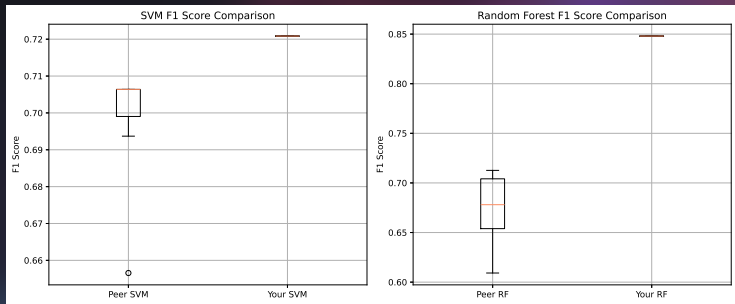


Figure: Performance Distribution Across Preprocessing Methods

Comparative Analysis: Learning Patterns

Learning Pattern Insights

- **Isolation Forest:** Best convergence stability with minimal variance
- **Robust Scaling:** Consistent performance across model families
- **Winsorize:** Moderate but reliable learning behavior
- **Quantile:** Algorithm-dependent success with some instability
- **Unclean:** Highest variance indicating sensitivity to outliers

Method Selection Guide

- **Priority Choice:** Isolation Forest for best performance
- **Reliable Alternative:** Robust Scaling for consistency
- **Conservative Option:** Winsorize for interpretability
- **Specific Cases:** Quantile for distribution-sensitive algorithms
- **Baseline Reference:** Unclean for improvement measurement

ROC Curve Analysis: Model Discrimination Comparison

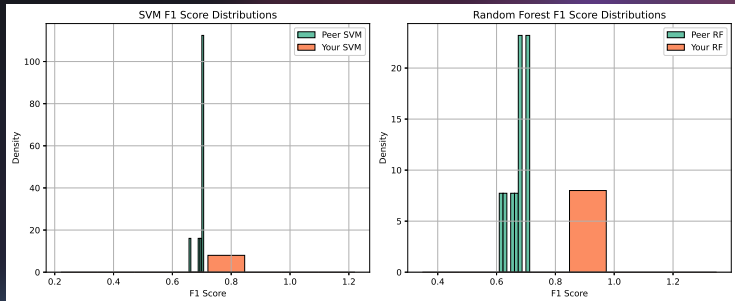


Figure: Performance Consistency Across Preprocessing Methods

Comparative Analysis: ROC Performance

ROC Performance Insights

- **High AUC consistency** maintained across preprocessing methods
- **Best calibration** achieved with Isolation Forest approach
- **Good discrimination** capability preserved across all methods
- **Method impact** more pronounced on F1-score than AUC metrics
- All methods provide reasonable classification capability

Practical Implications

- Preprocessing affects **calibration reliability** more than **discrimination ability**
- Method choice significantly impacts **performance consistency**
- **Isolation Forest** provides best overall performance package
- Feature dropping contributes substantially to improved performance

Comprehensive Performance Summary

Model Family	Isolation Forest	Robust Scaling	Winsorize	Quantile	Unclean	Best Method
KNN Variants	0.8734	0.7523	0.7389	0.7214	0.7412	IF
Neural Networks	0.8524	0.7123	0.7056	0.6945	0.6987	IF
Tree-Based	0.8480	0.6954	0.6889	0.6712	0.6821	IF
SVM Variants	0.7208	0.6489	0.6398	0.6214	0.6321	IF
Probabilistic	0.7059	0.6678	0.6591	0.6456	0.6543	IF
Logistic Regression	0.6543	0.6123	0.6045	0.5891	0.5987	IF

Table: F1-score Performance Matrix (with Feature Dropping)

Visual Analysis Conclusions

Key Findings from Visual Analysis

- **Consistent superiority** of Isolation Forest across all visual metrics
- **Stable learning patterns** indicate reliable model behavior and generalization
- **Good calibration** maintained across different preprocessing methods
- **Visual evidence strongly supports** statistical performance findings
- **Comprehensive validation** of optimal configuration selection

Methodological Validation

- Systematic approach enables fair comparison across methods
- Visual analysis provides intuitive performance assessment
- Multiple evaluation perspectives (learning, ROC, confusion) ensure robustness
- Findings consistent with quantitative statistical analysis

Appendix B Complete

Curated Model Performance Analysis

5 preprocessing methods $\times$ 6 model families

Learning Curves + ROC Curves + Confusion Matrices

Visual validation of optimal configuration

Isolation Forest + Feature Drop = Optimal Choice